Experiment 7

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1. Aim: To implement and analyze different types of SQL joins in PostgreSQL including INNER JOIN, LEFT JOIN, RIGHT JOIN, SELF JOIN, FULL JOIN, and CROSS JOIN.

2. Objective: To understand and apply various join operations on a relational database schema consisting of Students, Courses, Enrollments, and Departments tables to extract meaningful information.

3. Tools Used:

- PostgreSQL
- pgAdmin (for query execution)

4. Theory

In relational databases, data is often stored in multiple tables to maintain normalization. SQL joins are used to combine these tables based on related columns.

Different types of joins include:

- INNER JOIN: Returns only matching records from both tables.
- LEFT JOIN: Returns all records from the left table and matched records from the right table; otherwise NULL.
- RIGHT JOIN: Returns all records from the right table and matched records from the left table; otherwise NULL.
- FULL JOIN: Combines results of both LEFT and RIGHT joins, showing all records with NULLs where no match exists.
- CROSS JOIN: Produces all possible combinations (Cartesian product) of rows.
- SELF JOIN: A table joined with itself to compare rows within the same table.

These joins help in efficient querying, minimizing redundancy, and extracting related data.

5. Practical / Experiment Steps

Step 0: Table Creation

```
CREATE TABLE students (
    student_id INT PRIMARY KEY,
    student_name VARCHAR(50),
    dept_id INT
);
CREATE TABLE courses (
    course_id INT PRIMARY KEY,
    course_title VARCHAR(50)
);
CREATE TABLE enrollments (
    student_id INT,
    course_id INT,
    PRIMARY KEY (student_id, course_id)
);
CREATE TABLE departments (
    dept_id INT PRIMARY KEY,
    dept_name VARCHAR(50)
);
```

Step 1: Insert Sample Data

```
    -- Students
    INSERT INTO students VALUES
    (101, 'Aman', 1),
    (102, 'Riya', 2),
    (103, 'Rahul', 3),
    (104, 'Sneha', 4),
    (105, 'Arjun', 1);
    -- Courses
    INSERT INTO courses VALUES
    (201, 'Database Systems'),
    (202, 'Cloud Computing'),
```

```
(203, 'Artificial Intelligence'),
(204, 'Computer Networks');
-- Enrollments
INSERT INTO enrollments VALUES
(101, 201),
(101, 202),
(102, 203),
(103, 201),
(104, 204);
-- Departments
INSERT INTO departments VALUES
(1, 'Oracle'),
(2, 'SAP'),
(3, 'IBM'),
(4, 'Microsoft');
```

Step 2: INNER JOIN

```
SELECT s.student_id, s.student_name, c.course_title
FROM students s
INNER JOIN enrollments e ON s.student_id = e.student_id
INNER JOIN courses c ON e.course_id = c.course_id;
```

	student_id integer	student_name character varying (50)	course_title character varying (50)
1	101	Aman	Database Systems
2	101	Aman	Cloud Computing
3	102	Riya	Artificial Intelligence
4	103	Rahul	Database Systems
5	104	Sneha	Computer Networks

Step 3: LEFT JOIN

```
SELECT s.student_id, s.student_name
FROM students s
LEFT JOIN enrollments e ON s.student_id = e.student_id
WHERE e.student_id IS NULL;
```

	student_id [PK] integer	student_name character varying (50)
1	105	Arjun

Step 4: RIGHT JOIN

```
SELECT s.student_name, c.course_title
FROM enrollments e
RIGHT JOIN courses c ON e.course_id = c.course_id
LEFT JOIN students s ON e.student_id = s.student_id;
```

	student_name character varying (50)	course_title character varying (50)
1	Aman	Database Systems
2	Aman	Cloud Computing
3	Riya	Artificial Intelligence
4	Rahul	Database Systems
5	Sneha	Computer Networks

Step 5: Join wth departments

```
SELECT s.student_id, s.student_name, d.dept_name
FROM students s
JOIN departments d ON s.dept_id = d.dept_id;
```

	student_id integer	student_name character varying (50)	dept_name character varying (50)
1	101	Aman	Oracle
2	102	Riya	SAP
3	103	Rahul	IBM
4	104	Sneha	Microsoft
5	105	Arjun	Oracle

Step 6: CROSS JOIN

```
SELECT s.student_name, c.course_title
FROM students s
```

```
CROSS JOIN courses c;
```

	student_name character varying (50)	course_title character varying (50)
1	Aman	Database Systems
2	Riya	Database Systems
3	Rahul	Database Systems
4	Sneha	Database Systems
5	Arjun	Database Systems
6	Aman	Cloud Computing
7	Riya	Cloud Computing
8	Rahul	Cloud Computing
9	Sneha	Cloud Computing
10	Arjun	Cloud Computing
11	Aman	Artificial Intelligence
12	Riya	Artificial Intelligence
13	Rahul	Artificial Intelligence
14	Sneha	Artificial Intelligence
15	Arjun	Artificial Intelligence
16	Aman	Computer Networks
17	Riya	Computer Networks
18	Rahul	Computer Networks
19	Sneha	Computer Networks
20	Arjun	Computer Networks

Step 7: FULL JOIN

```
SELECT s.student_name, c.course_title
FROM students s
FULL JOIN enrollments e ON s.student_id = e.student_id
FULL JOIN courses c ON e.course_id = c.course_id;
```

	student_name character varying (50)	course_title character varying (50)
1	Aman	Database Systems
2	Aman	Cloud Computing
3	Riya	Artificial Intelligence
4	Rahul	Database Systems
5	Sneha	Computer Networks
6	Arjun	[null]

6. Learning Outcome:

- Gained practical understanding of SQL joins and their real-world applications.
- Learned to retrieve and combine data from multiple related tables.
- Understood differences between INNER, LEFT, RIGHT, FULL, and CROSS joins.
- Developed ability to write efficient queries for data analysis.
- Improved hands-on experience with PostgreSQL.